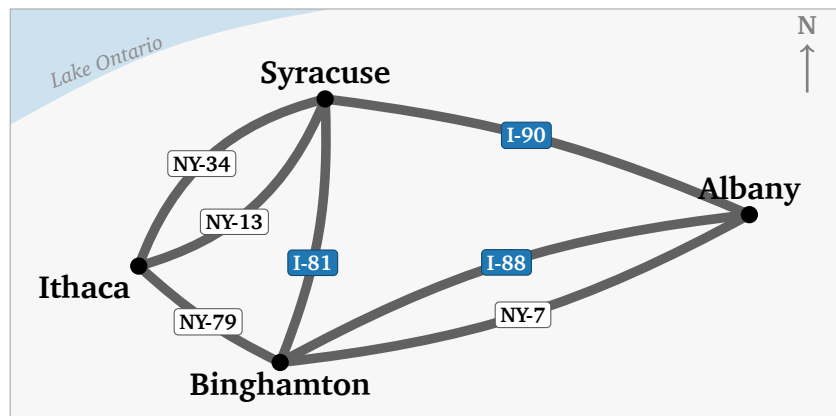


Drive Every Highway Once, Then Teach a Computer to Do It

Do this before the lecture, in pencil. Nothing here needs a word you have not been given on the sheet.

Part 1 — Seven highways

Seven highways join four Upstate New York cities: Ithaca, Syracuse, Binghamton, Albany. Drive every highway exactly once.



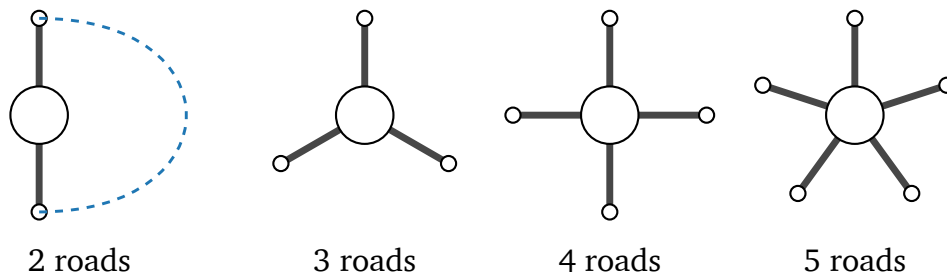
Question 1(a): Find such a route. Start and finish anywhere. Cities may repeat; highways may not. Write a city as its first letter and a highway as its number. Two steps are done for you.

I 13 S 34 I _____

Question 1(b): Draw an eighth highway on the map: **US-11**, Binghamton to Syracuse, alongside I-81. Try again. Which highway is left over?

Question 2: One city, on its own. Arriving uses up one highway, leaving uses up another — so highways get used in **pairs**.

Join two road-ends with a curved line to mean “in here, out there”. Pair up as far as you can. The first is done.



	2 roads	3 roads	4 roads	5 roads
pairs you drew	1			
roads left over	0			

- (a) Which ones can you drive straight *through*? _____
- (b) What do those numbers share? _____
- (c) A left-over road has no partner, so this city must be the tour's _____

Question 3: Count the highways at each city. Ithaca is done.

	I	S	B	A	how many have a left-over?
seven roads	3				
with US-11 too					

One start, one finish. So at most _____ cities may have a left-over road.

Seven roads: _____ Eight roads: _____

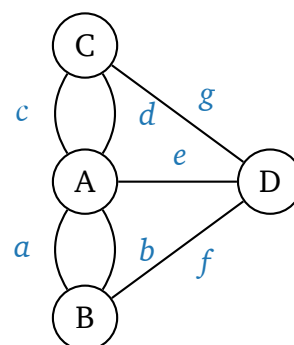
No route with eight because _____

Question 4: The same puzzle, 1736. Königsberg: four pieces of land, seven bridges. Two pairs join the same land; both of a pair must be crossed.

Bridges: A _____ B _____ C _____ D _____

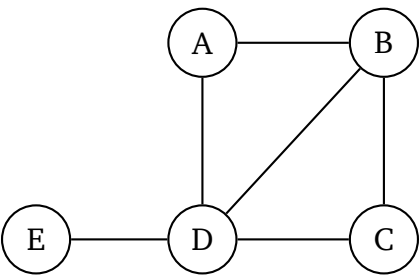
Is the walk possible? _____

Destroying one bridge is enough. Which ones work? _____



Part 2 — Three kinds of route

A different network: five bus stops.



Question 5(a): Fill in the table. First row done.

route	repeats a line?	repeats a stop?	name
A B D B C	yes (B–D twice)	yes (B)	
A B D C B			
A B C D E			

Question 5(b): Write the right name in the last column.

Repeats nothing: **path**. Never repeats a line: **trail**. Repeats both: **walk**.

Question 5(c): A route from A to E that repeats a stop but no line:

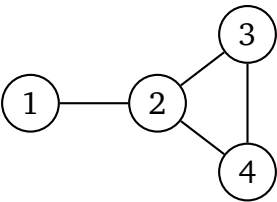
A _____ E

Question 5(d): Your route in 1(a) is a _____.

Question 5(e): It cannot be a path: 7 lines need _____ different cities, and the map has _____.

Part 3 — Predictions for the lab

Fill this in on paper first, then check it in the lab. A new network:



Question 6(a): Write **1** in row i , column j if a line joins i and j , else **0**. Row 1 done.

$$A = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 1 & 0 & 0 \\ 2 & & & & \\ 3 & & & & \\ 4 & & & & \end{array}$$

Question 6(b): Add up each row, then compare with the picture.

row totals: _____

Each row total is the _____ of that node.

Question 6(c): A **2-step route** is one line, then one more. Count them on the picture.

1 to 3: how many? _____ list them: $1 \rightarrow _____ \rightarrow 3$

1 to 2: how many? _____ why? _____

2 back to 2: how many? _____

Question 6(d): The lab will square your grid. What do you expect in A^2 ?

row 1, column 3: _____ row 1, column 2: _____

the diagonal: _____

Now the lab, in pairs. Bring this sheet. One laptop, two people: the one not typing says the answer before the other one runs the cell.

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